Skills Review

Exercise No. 15

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Course/Section: CPE4B **Date: Oct. 7, 2026**

Machine Problem:

Create a program using Interrupt Function 10h that will display a chess board with any combination of two colors that you want. Graph the coordinates on the figure below and shade each square of differing colors. Save the program using Exer15.asm as the filename.

Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B
; Exer15 - Chess Board (Interrupt 10h)
; each square = 2 rows x 4 columns, board = rows 4-19, columns 24-55

.model small
.stack
.data

var1    DB 'CHESS BOARD','$'
var2    DB '8','$'
var3    DB '7','$'
var4    DB '6','$'
var5    DB '5','$'
var6    DB '4','$'
var7    DB '3','$'
var8    DB '2','$'
var9    DB '1','$'
var10   DB 'a   b   c   d   e   f   g   h','$'
var11   DB 'Light squares: WHITE      Dark squares: GREEN','$'

.code
start:

        ; screen background (0,0 - 24,79)
        mov ah, 06h             ; Function 06h - scroll up window
        mov bh, 1Eh             ; blue background, yellow text
        mov ch, 0               ; start row
        mov cl, 0               ; start column
```

```
        mov dh, 24              ; end row
        mov dl, 79              ; end column
        int 10h                 ; BIOS video interrupt - display the window

        ; board frame (3,22 - 20,57)
        mov ah, 06h
        mov bh, 6Eh             ; yellow background, light yellow text
        mov ch, 3
        mov cl, 22
        mov dh, 20
        mov dl, 57
        int 10h

        ; whole board - light squares (4,24 - 19,55)
        mov ah, 06h
        mov bh, 70h             ; white
        mov ch, 4
        mov cl, 24
        mov dh, 19
        mov dl, 55
        int 10h

        ; <---- dark squares (green) ---->
        ; <---- rank 8 ---->
        ; b8 (4,28 - 5,31)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 4
        mov cl, 28
        mov dh, 5
        mov dl, 31
        int 10h

        ; d8 (4,36 - 5,39)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 4
        mov cl, 36
        mov dh, 5
        mov dl, 39
```

```
        int 10h

        ; f8 (4,44 - 5,47)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 4
        mov cl, 44
        mov dh, 5
        mov dl, 47
        int 10h

        ; h8 (4,52 - 5,55)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 4
        mov cl, 52
        mov dh, 5
        mov dl, 55
        int 10h

        ; <---- rank 7 ---->
        ; a7 (6,24 - 7,27)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 6
        mov cl, 24
        mov dh, 7
        mov dl, 27
        int 10h

        ; c7 (6,32 - 7,35)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 6
        mov cl, 32
        mov dh, 7
        mov dl, 35
        int 10h

        ; e7 (6,40 - 7,43)
```

```
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 6
        mov cl, 40
        mov dh, 7
        mov dl, 43
        int 10h

        ; g7 (6,48 - 7,51)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 6
        mov cl, 48
        mov dh, 7
        mov dl, 51
        int 10h

        ; <---- rank 6 ---->
        ; b6 (8,28 - 9,31)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 8
        mov cl, 28
        mov dh, 9
        mov dl, 31
        int 10h

        ; d6 (8,36 - 9,39)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 8
        mov cl, 36
        mov dh, 9
        mov dl, 39
        int 10h

        ; f6 (8,44 - 9,47)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 8
```

```
        mov cl, 44
        mov dh, 9
        mov dl, 47
        int 10h

        ; h6 (8,52 - 9,55)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 8
        mov cl, 52
        mov dh, 9
        mov dl, 55
        int 10h

        ; <---- rank 5 ---->
        ; a5 (10,24 - 11,27)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 10
        mov cl, 24
        mov dh, 11
        mov dl, 27
        int 10h

        ; c5 (10,32 - 11,35)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 10
        mov cl, 32
        mov dh, 11
        mov dl, 35
        int 10h

        ; e5 (10,40 - 11,43)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 10
        mov cl, 40
        mov dh, 11
        mov dl, 43
```

```
        int 10h

        ; g5 (10,48 - 11,51)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 10
        mov cl, 48
        mov dh, 11
        mov dl, 51
        int 10h

        ; <---- rank 4 ---->
        ; b4 (12,28 - 13,31)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 12
        mov cl, 28
        mov dh, 13
        mov dl, 31
        int 10h

        ; d4 (12,36 - 13,39)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 12
        mov cl, 36
        mov dh, 13
        mov dl, 39
        int 10h

        ; f4 (12,44 - 13,47)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 12
        mov cl, 44
        mov dh, 13
        mov dl, 47
        int 10h

        ; h4 (12,52 - 13,55)
```

```
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 12
        mov cl, 52
        mov dh, 13
        mov dl, 55
        int 10h

        ; <---- rank 3 ---->
        ; a3 (14,24 - 15,27)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 14
        mov cl, 24
        mov dh, 15
        mov dl, 27
        int 10h

        ; c3 (14,32 - 15,35)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 14
        mov cl, 32
        mov dh, 15
        mov dl, 35
        int 10h

        ; e3 (14,40 - 15,43)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 14
        mov cl, 40
        mov dh, 15
        mov dl, 43
        int 10h

        ; g3 (14,48 - 15,51)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 14
```

```
        mov cl, 48
        mov dh, 15
        mov dl, 51
        int 10h

        ; <---- rank 2 ---->
        ; b2 (16,28 - 17,31)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 16
        mov cl, 28
        mov dh, 17
        mov dl, 31
        int 10h

        ; d2 (16,36 - 17,39)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 16
        mov cl, 36
        mov dh, 17
        mov dl, 39
        int 10h

        ; f2 (16,44 - 17,47)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 16
        mov cl, 44
        mov dh, 17
        mov dl, 47
        int 10h

        ; h2 (16,52 - 17,55)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 16
        mov cl, 52
        mov dh, 17
        mov dl, 55
```

```
        int 10h

        ; <---- rank 1 ---->
        ; a1 (18,24 - 19,27)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 18
        mov cl, 24
        mov dh, 19
        mov dl, 27
        int 10h

        ; c1 (18,32 - 19,35)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 18
        mov cl, 32
        mov dh, 19
        mov dl, 35
        int 10h

        ; e1 (18,40 - 19,43)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 18
        mov cl, 40
        mov dh, 19
        mov dl, 43
        int 10h

        ; g1 (18,48 - 19,51)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 18
        mov cl, 48
        mov dh, 19
        mov dl, 51
        int 10h

        mov ax, @data           ; initialize the data segment
```

```
        mov ds, ax

        mov ah, 02h             ; Function 02h - set cursor position
        mov bh, 00h             ; page 0
        mov dh, 1               ; row
        mov dl, 34              ; column
        int 10h
        mov ah, 09h             ; Function 09h - display string
        mov dx, offset var1     ; title
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 4
        mov dl, 23
        int 10h
        mov ah, 09h
        mov dx, offset var2     ; rank 8
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 6
        mov dl, 23
        int 10h
        mov ah, 09h
        mov dx, offset var3     ; rank 7
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 8
        mov dl, 23
        int 10h
        mov ah, 09h
        mov dx, offset var4     ; rank 6
        int 21h

        mov ah, 02h
        mov bh, 00h
```

```
        mov dh, 10
        mov dl, 23
        int 10h
        mov ah, 09h
        mov dx, offset var5    ; rank 5
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 12
        mov dl, 23
        int 10h
        mov ah, 09h
        mov dx, offset var6    ; rank 4
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 14
        mov dl, 23
        int 10h
        mov ah, 09h
        mov dx, offset var7    ; rank 3
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 16
        mov dl, 23
        int 10h
        mov ah, 09h
        mov dx, offset var8    ; rank 2
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 18
        mov dl, 23
        int 10h
        mov ah, 09h
```

```
        mov dx, offset var9     ; rank 1
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 20
        mov dl, 25
        int 10h
        mov ah, 09h
        mov dx, offset var10    ; file letters a-h
        int 21h

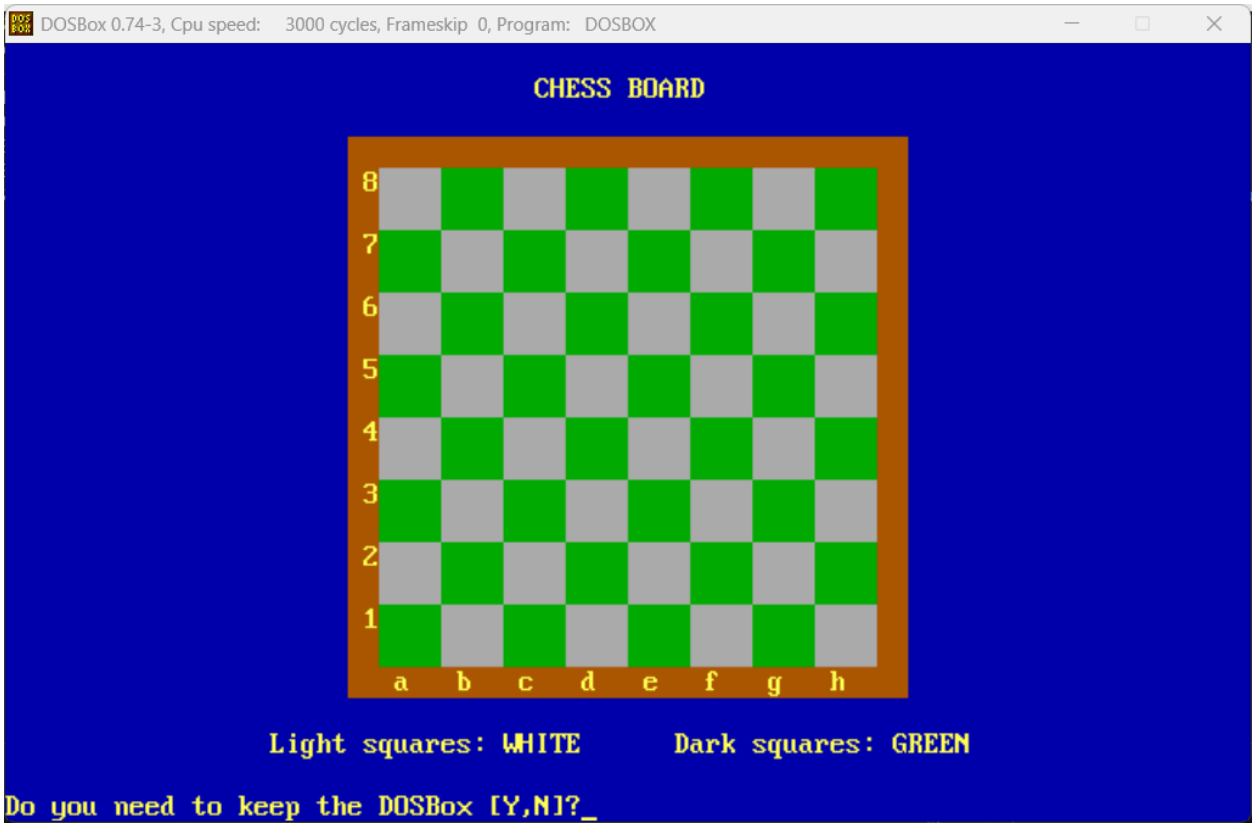
        mov ah, 02h
        mov bh, 00h
        mov dh, 22
        mov dl, 17
        int 10h
        mov ah, 09h
        mov dx, offset var11    ; color legend
        int 21h

        mov ah, 02h             ; move the cursor near the bottom so the
        mov bh, 00h             ; DOS prompt will not cover the design
        mov dh, 23
        mov dl, 0
        int 10h

        mov ah, 4Ch             ; terminate program
        int 21h

end start
```

Output:


DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX
CHESS BOARD
8
7
6
5
4
3
2
1
a b c d e f g h
Light squares: WHITE
Dark squares: GREEN
Do you need to keep the DOSBox [Y,N]?_